SE3002 Assignment 2 — Testing Report

Group: Ghilman Shad Ali, Mohammad Ramzan, Atiq-ur-Rehman **Repository:** https://github.com/GhilmanShad/SQE_Assignment_2

1. Black-box design (Part A)

- Expected results derived from REQUIREMENTS.md only. Cases live in one registry in `tests/test_blackbox.py`; spreadsheets in `docs/blackbox_test_design.xlsx` and `docs/decision_table.xlsx` are generated from it.
- **ECP:** 21 cases (credit hours, dues, prerequisites, availability, validation). **BVA:** 29 cases (credits 0/18, course credits 1/4, 18-credit total, capacity/enrolled, dues 0/1). **Decision table:** 6 outcome rules + 2 invalid-input rules, 14 DT tests including priority pairs (D>R>P>C>S).
- **Optimisation (A5):** 64 technique IDs map to 53 executed tests; 11 exact duplicates (same inputs + requirement + expected) were merged, keeping all IDs in the test id (e.g. `ECP-06+ECP-12+BVA-12+DT-06`). Cases with equal outputs but different inputs/objectives (e.g. `BVA-11` dues=1 vs `ECP-05` dues=5000) were deliberately kept.
- **Initial result:** 93 black-box tests: 74 passed, 19 failed (`docs/initial_blackbox_run.txt`).

2. White-box analysis (Part B)

CFG (`docs/cfg.png`) of `registration_decision()`.

Convention	N	E	V(G)=E−N+2	decisions+1
Compound `not 0<=e<=cap or cap<1` = 1 predicate	16	22	8	7+1 = 8
Each atomic condition = own node (A, B, C)	18	26	**10**	9+1 = 10

Basis paths (10): P01 dues<0 · P02 A false · P03 A true,B false · P04 A,B true,C true · P05 DUES_BLOCK · P06 ALREADY · P07 PREREQ · P08 CREDIT · P09 WAITLISTED · P10 REGISTERED. Before the fix, **P09 was infeasible** because line 48 already rejected enrolled>capacity, so the old `enrolled > capacity` check

at line 58 was dead code (D4). After the D4 fix (`enrolled == capacity`), P09 is reachable and is covered by `test_WB_P09_full_course_waitlisted` plus the black-box full-course cases. Tests: `test_WB_P01…P10`.

MC/DC, line 48 `not (A and B) or C` (A: 0<=enrolled, B: enrolled<=capacity, C: capacity<1), Python short-circuit shown as NE = not evaluated:

Test	A	B	C	Decision	Data
R1	T	T	F	False	enrolled 10, cap 30
R2	T	T	T	True	enrolled 0, cap 0
R3	T	F	NE	True	enrolled 31, cap 30
R4	F	NE	NE	True	enrolled −1, cap 30
Independence pairs: **A (R1,R4)**, **B (R1,R3)**, **C (R1,R2)**. Also done: lines 30/32 (3 tests each) and `register_course` line 70 (`X and Y`: pairs X (T1,T3), Y (T1,T2)).					

Coverage (pre-fix stages measured with a local tracer; the after-fix stage re-measured with `pytest-cov`):

Stage	Statement	Branch	Uncovered
Black-box only	96.4% (53/55)	92.3% (24/26)	line 59 `return "WAITLISTED"`; line 71 waitlist append
+ white-box (`tests/test_whitebox.py`)	98.2% (54/55)	96.2% (25/26)	line 59 — unreachable until D4 is fixed

Stage	Statement	Branch	Uncovered
After D1–D5 fixes	**100%** (54/54)	**100%** (26/26)	none
Line 71 was covered by `test_WB_MCDC70_T1` using `mock.patch` on `registration_decision`. The after-fix figures come from `python -m pytest tests/ -v --cov=registration --cov-branch --cov-report=term-missing` (122 passed).			

3. Defects (see docs/DEFECTS.md)

D1 credit `<18` · D2 prerequisites any-vs-all · D3 dues threshold 1000 · D4 waitlist condition `>` (dead code) · D5 withdraw doesn't decrement enrolled. Issues: #5 (D1), #6 (D2), #7 (D3), #8 (D4), #9 (D5). Fix PR: #10 (merged). After fixes: **122/122 tests pass**.

4. GitHub Evidence and Contributions

Repository: https://github.com/GhilmanShad/SQE_Assignment_2 **Workflow:** tests.yml — runs `pytest --cov` on every push and PR and uploads `test-results.xml`, `coverage.xml`, and `htmlcov/` as the `testing-evidence` artifact.

- Pre-fix red run (19 failed, 83 passed): 43481845845
- Post-fix green run on `main` (122 passed, 100% statement/branch coverage): 43483968026

Pull requests: #1 black-box tests (merged) · #3 white-box tests and report (merged) · #10 defect fixes (merged).

Group Members and Contributions

Group Member	Contributions
Ghilman Shad Ali	Black-Box Testing
Mohammad Ramzan	GitHub Workflows

Group Member	Contributions
Atiq-ur-Rehman	White-Box Testing